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# Smart Urban Waste Management

Using Truck-Mounted Cameras to Clean Cities Better

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## **What is this document?**

A garbage truck drives around the city with a camera on it. This document explores how we can turn that simple setup into a system that makes city cleaning smarter, cheaper, and more accountable.

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## Why This Problem Is Of Interest

Before any architecture or metric, there is a simpler motivation: we live in Morocco, and we encounter this problem every day. An overflowing bin is not an exceptional event in our neighbourhoods — it is background noise. Waste piled on the pavement, bags torn open overnight, the smell of organic matter on a hot afternoon. We have learned, almost without noticing, to walk past it.

That normalisation is itself the most insidious part of the problem. When a bin overflows every week without consequence, residents stop reporting it, the municipality stops hearing about it, and the cleaning company faces no pressure to change. The street stays dirty — not through neglect exactly, but through the absence of any feedback loop.

## What It Actually Costs

The consequences extend well beyond the aesthetic:

- **A health hazard at ground level.** In dense residential neighbourhoods, the street is where children play. Broken glass, sharp metal lids, and rotting organic matter around an overflowing bin become part of that space. This is not a theoretical risk — it is a recurring condition in many Moroccan cities.
- **Overflow that compounds itself.** Stray animals tear into bags left on the ground. What begins as a localised overflow scatters across a wide radius overnight. By the time a truck arrives, it has become a manual cleanup operation rather than a routine collection — slower, more expensive, and entirely avoidable.
- **Drainage failure and flooding.** Waste accumulating near gutters blocks water flow, particularly after rain. In older neighbourhoods with limited drainage infrastructure, this directly contributes to street flooding — a secondary hazard born from an unresolved primary one.
- **An unequal burden by district.** The problem does not fall evenly across the city. Denser, lower-income areas — already underserved in bin density and collection frequency — bear the most visible impact. Wealthier districts generate complaints and receive faster responses; others have learned not to expect them. The current system has no mechanism to detect, let alone correct, this gap.

### Local knowledge that exists nowhere official.

We know which corners of our cities are reliably dirty. Which market days leave certain streets buried in waste by evening. Which specific bins overflow every single week. This is granular, lived knowledge — and it resides entirely in the heads of residents, invisible to any official system. One of the things we are trying to build is exactly the infrastructure to capture it.

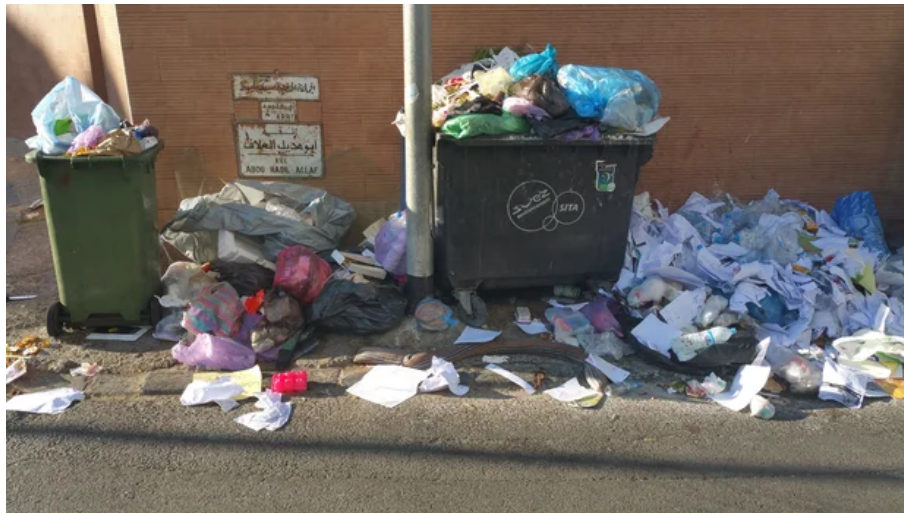


Figure 1: Example of an Overflowing Trash Bin

**The link to the system we are proposing.**

Every problem described above has a root cause that our system directly addresses: **nobody currently knows the state of municipal bins in real time.**

The overflowing bin near the school exists because no one was notified it was full. The scattered waste exists because no one dispatched a truck before animals got to it. The unequal service exists because there is no data to show municipalities which zones are underserved.

A camera on a garbage truck, with the right system behind it, closes that gap. It makes the invisible visible. And it gives the people responsible for urban cleanliness a tool that actually matches the scale of the problem.

# The Big Picture

## What We Have

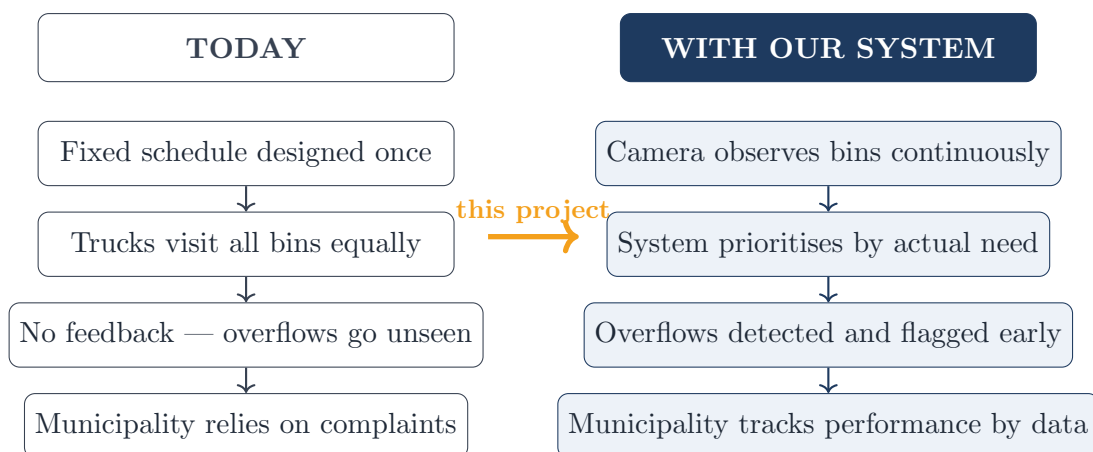
A fleet of garbage trucks roams the city daily, each fitted with a forward-facing camera. As the trucks follow their routes, the camera captures continuous video of the streets, bins, and surrounding environment. Every frame is paired with a GPS coordinate and a timestamp from the truck's on-board system. This gives us two things at once: **what** the city looks like at a given moment, and **where and when** that observation was made. That combination — visual data anchored in space and time — is the raw material the entire system is built on.

## What We Are Trying to Do

Today, city cleaning operates on **fixed schedules**: trucks visit every street and every bin at pre-set times, regardless of whether those bins are full, half-empty, or overflowing. There is no live view of the city's actual cleanliness state. Routes are planned once and revised rarely. The system is, by design, blind to reality.

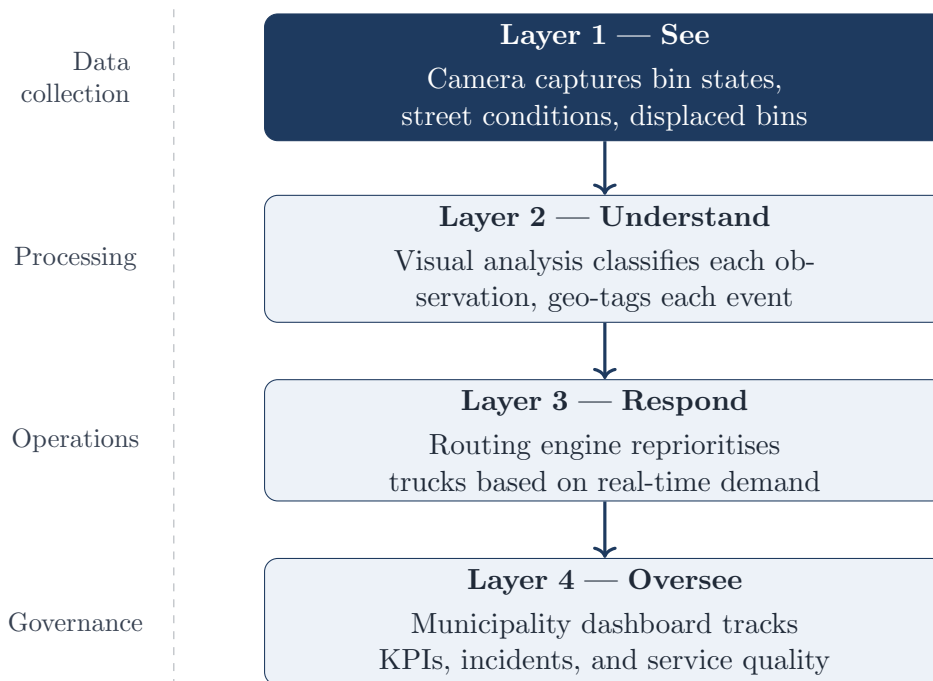
**Analogy.** Imagine watering every plant in your garden on the same schedule, every day, regardless of soil moisture, rainfall, or season. Some plants drown. Others dry out. You are not managing the garden — you are running a timer. That is how most cities manage waste collection today. What we want to build is the soil sensor: a system that tells you which plants actually need water, and when.

The goal is to shift from **reactive** operation — *we come because the schedule says so* — to **predictive and adaptive** operation — *we come because the data tells us we are needed*. The diagram below illustrates where this project sits relative to how things work today.



## The Pipeline in One View

The system we are proposing is not a single tool — it is a chain of four connected layers, each feeding the next. Understanding this structure is essential before looking at any individual problem.



## Three Problems, One System

The pipeline above is our answer to three distinct but connected failures in how cities manage waste today. Each one is explored in its own section — but it helps to name them here first, so the structure is clear from the outset.

- **We don't know what the streets look like.** There is no live, city-wide picture of bin states or street cleanliness between collection rounds. Overflows go undetected. Displaced bins go unreported. The city is effectively invisible to the people managing it.
- **Truck routes are blind to actual need.** Collection schedules are fixed and rarely revised. Trucks visit bins that are nearly empty while others overflow. The system spends resources in the wrong place, at the wrong time, by design.
- **The municipality has no objective feedback.** There is no continuous, data-driven record of how well the cleaning service is performing. Oversight relies on complaints and self-reporting — neither of which is reliable, and neither of which scales.

These three problems are not independent. The first makes the second worse (you cannot route intelligently without observation), and both make the third impossible (you cannot hold a service accountable without data). Solving them together, as a layered system, is precisely the point.

## Problem 1 — We Don't Know What the Streets Look Like

### The Problem

Between collection rounds, no one has a reliable picture of what the city actually looks like. A bin may have been overflowing since yesterday morning. Another may have been knocked over near the market. An illegal dump may be forming behind the bus station. None of this is known until someone physically sees it — and even then, it is rarely recorded.

The three ways cities currently find out about problems are:

- A citizen complaint — slow, patchy, and skewed toward wealthier areas
- A manual inspection — expensive and too infrequent to matter
- A truck driver noticing in passing — informal and undocumented

All three are reactive. None produce structured, reusable data.

### What the Camera Changes

As the truck drives its route, the camera passively captures everything it passes. With the right processing layer, each frame becomes a structured observation: fill level, bin position, waste on the ground, anomalies. Every observation is timestamped and GPS-tagged automatically.



Accumulated over time, these observations become a **live cleanliness map of the city** — a continuously updated record of the state of every bin and street segment the trucks pass.

**Overflow near a school.** The truck passes a bin at 8:04 AM on Monday. The camera detects waste above the rim and logs it: *Bin #0472, Avenue Hassan II — overflow*. A supervisor is notified and dispatches a vehicle before school drop-off. Without this system, the bin waits until Wednesday's scheduled round.

**Recurring displacement.** A bin near a weekly souk is knocked over every Saturday night. After three occurrences, the system flags a pattern. The municipality can now act on evidence rather than coincidence: anchor the bin, add a weekend pass, or relocate it.

### What This Produces

A continuously updated **city cleanliness map**. This is the data foundation the next two solutions are built on.

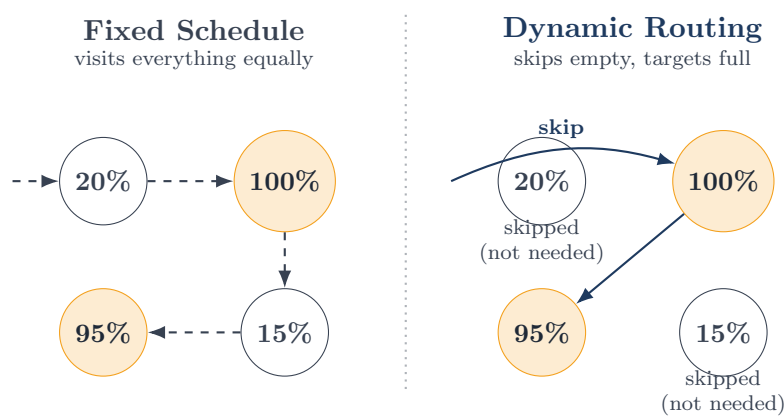
## Problem 2 — Truck Routes Are Blind to Actual Need

### The Problem

Collection routes are designed once and rarely revised. Every bin is visited on the same cycle, regardless of how full it actually is. This creates two simultaneous failures:

- **Over-collection:** trucks empty bins that are barely a quarter full, wasting fuel, time, and vehicle wear.
- **Under-collection:** high-footfall bins at markets, bus stops, and tourist zones overflow before the next scheduled visit.

The diagram below shows the difference in practice:



**Same schedule, opposite problems.** A quiet residential bin is collected Monday and Thursday — averaging 30% full each time. A bin outside the central market overflows every Tuesday afternoon. Both are on identical schedules. The system is deployed where it is least needed.

### What Fill Data and GPS Unlock

Combining camera observations with fleet GPS gives us **demand data**: which bins fill fastest, which zones are consistently high-load, and where trucks are covering ground without finding much to collect.

This feeds a **dynamic routing layer** that replaces fixed circuits with demand-driven priorities. Each morning, the system proposes a ranked collection plan based on observed fill levels and historical patterns.

**Market day adaptation.** Bins near the medina souk reliably reach 90% capacity by 2 PM every Sunday. The system schedules an additional Sunday afternoon pass for that zone and removes a redundant Wednesday visit. Fewer overflows, less distance driven.

Aggregated across the full fleet, the same data enables **real-time load balancing**: if one truck finishes its priority bins ahead of schedule, the system can redirect it to assist an overloaded route.

## Problem 3 — The Municipality Has No Objective Feedback

### The Problem

The municipality pays the cleaning company. But there is currently no objective way to know whether the service is working. Oversight depends on:

- Citizen complaints — reactive, emotionally biased, uneven by district
- Periodic manual audits — expensive, infrequent, not statistically robust
- Self-reporting from the contractor — an obvious conflict of interest

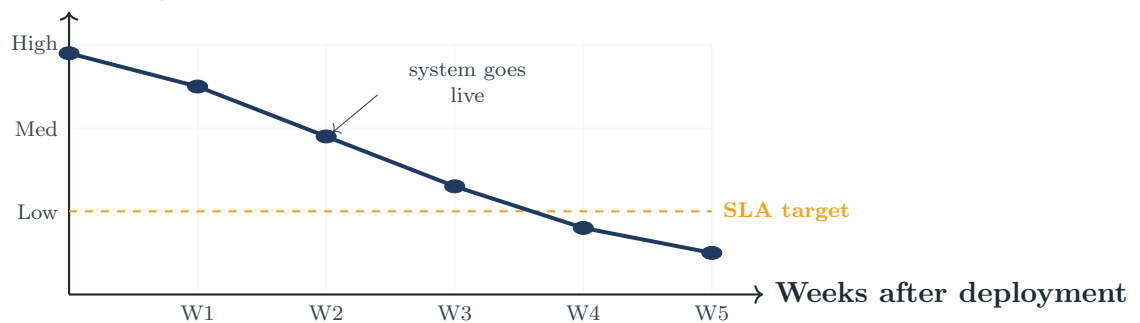
Without a continuous, georeferenced record of city cleanliness, it is impossible to assess whether performance is improving, identify chronic problem zones, or evaluate whether any intervention actually helped.

### What the Data Layer Provides

Every observation the camera system produces — bin state, overflow event, illegal dump, displacement — is timestamped, geolocated, and logged. Accumulated over weeks and months, this becomes an objective performance record.

It feeds a **municipality dashboard**: cleanliness trends by district, incident rates by bin, and response times by zone — updated continuously, without relying on anyone to self-report.

Overflow incidents / week

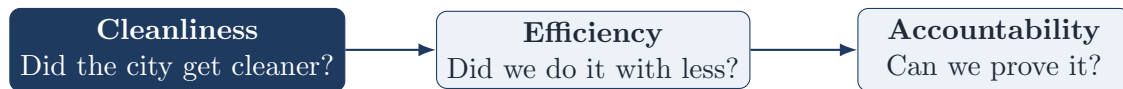


**District benchmarking.** The dashboard shows District 3 averaging 4.2 overflow incidents per week, while District 1 — same population, similar bin density — averages 0.8. The municipality can now ask a precise question: is this a routing gap, a bin density problem, or a behavioural pattern? The data makes it answerable.



## How We Measure Success — KPIs

Three categories of metrics map directly onto the three problems. Each one is only meaningful in relation to the others.



### Cleanliness

- **Overflow Rate**  
Overflow incidents detected per zone per week. The primary signal that the system is working. *Target:  $-X\%$  within 6 months of deployment.*
- **Average Fill Level at Collection**  
The healthy range is 70–90%: below that, trucks are arriving unnecessarily; above it, we are already too late.
- **Time-to-Resolution**  
Time elapsed between anomaly detection and confirmed resolution. *Target: under 4 hours during operational hours.*
- **Street Cleanliness Score**  
A 1–5 rating per street segment derived from camera observations, aggregated weekly across the city.

### Efficiency

- **Kilometres per Tonne Collected**  
Total distance driven divided by waste collected. The core routing efficiency metric — lower is better.
- **Empty Collection Rate**  
Percentage of bin visits where fill level was below 20%. Directly quantifies wasted trips.
- **Fleet Utilisation**  
Average truck load at return to facility. Tracks whether vehicles are making full use of each trip or returning half-empty.

### Accountability

- **Coverage Rate**  
Percentage of registered bins observed by a camera at least once per day. Ensures no zone falls into a blind spot.
- **Citizen Complaint Volume**  
Formal complaints logged with the municipality, tracked monthly by district.

External validation that the system's improvements are visible to residents.

### KPIs must be read as a set.

If the overflow rate drops 40% but kilometres per tonne *increases*, the system is preventing overflows by dispatching more trucks — not by routing smarter. The cleanliness metric improves while the efficiency metric regresses. That tension is exactly the kind of signal these KPIs are designed to surface.

## Next Steps

Before moving forward, we need to align on two things:

### Which Problem to Center

We could present all three problems as one integrated system (ambitious, coherent, but complex to scope). Or we could focus:

1. **Focus on Observation** — Build the camera detection pipeline and the cleanliness map. Simpler scope, immediately visible output.
2. **Focus on Routing** — Use GPS + fill data to optimize truck routes. Highest cost-reduction story; strongest business case for the company.
3. **Focus on the Municipality Dashboard** — Build the oversight layer. Most politically useful; positions us as a neutral accountability tool.
4. **Present all three as a unified system** — Strongest technically, but requires clear explanation of how layers connect.

### What Comes Next for Us

Once we agree on the above, we can then move to the **engineering design** part of this project:

Define the technical architecture: How the camera data is processed, What models are used, How routing is computed, and How data flows between components.

### Recommended path.

Our suggestion is to present all three layers as a unified system — because each layer's value depends on the others — but to emphasize **routing optimization** as the primary deliverable with the clearest measurable return. Detection enables it; the dashboard validates it.